

$$\begin{bmatrix} 2 & 3 & 6 \\ 4 & 6 & 3k \end{bmatrix}$$

$$\Rightarrow r_2 - 2r_1$$

$$\Rightarrow \begin{bmatrix} 2 & 3 & 6 \\ 0 & 0 & 3k-12 \end{bmatrix}$$

$$\Rightarrow \text{rank} = 1$$

$$k = 4$$

$\Rightarrow$ consistent,
infinite,

$\Rightarrow$

$$\text{rank} = 2$$

$\Rightarrow$

inconsistent

$$k \neq 4$$

no-soln